BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, DI KHAN

INTERMEDIATE (11TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021 • Paper Code: DIK-HSSC-XI-2021 (A)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – A

Time Allowed: 20 min | **Maximum Marks:** 20 | **Paper Code:** DIK-HSSC-XI-2021 (A)

Q1. Choose the correct option i.e. A,B,C,D and fill it up in the correct bubble in the given MCQs sheet.

Qi. The real and imaginary part in $(5 - 3i)^{-1}$ is _____

(A) x = 5/34, y = −3/34
(B) x = −5/34, y = −3/34
(C) x = 5/34, y = 3/34
(D) x = −5/34, y = 3/34

Qii. $(-1)^{-23/2}$ = _____

(A) −i
(B) i
(C) 1
(D) −1

Qiii. If A = [1 −2 1; 0 1 −2; −1 2 0] then A_{32} = _____

(A) −3
(B) −2
(C) 2
(D) 3

Qiv. If A = [3 2; 4 −1; 6 1] then A^t = _____

(A) [3 4 6; 2 1 −1]
(B) [3 4 6; 2 1 1]
(C) [3 4 6; 2 −1 −1]
(D) [3 4 6; 2 −1 1]

Qv. jxi = _____

(A) 0
(B) K
(C) −K
(D) i

Qvi. If a = 3i − 2i then â = _____

(A) (2i−3j)/√13
(B) (2i+3j)/√13
(C) (3i−2j)/√13
(D) (3i−2j)/√13

Qvii. Which of the following is not an arithmetic progression (A.P)?

(A) 2,4,8,16.....
(B) 3,6,9,12.....
(C) 3,7,11,15.....
(D) 4,8,12,16.....

Qviii. The first four terms of a sequence $(-1)^{n-1}\ 2n^{+1}$ is_____

(A) 4,8,16,32
(B) 4,−8,16,−32
(C) −4,8,−16,32
(D) 4,8,−16,32

Qix. Sum of the series to n terms 1.2+2.3+3.4+.....

(A) n(n+1)(n+2)/3
(B) (n+1)(n+2)/3
(C) (n+1)(n+2)/2
(D) n(n+1)(n+2)/2

Qx. $^{20}P_2$ = _____

(A) 190
(B) 380
(C) 660
(D) 760

Qxi. If A and B are mutually exclusive events, then _____

(A) P(A∩B) = P(A)+P(B)
(B) P(A∪B) = P(A)+P(B)
(C) P(A∪B) = P(A)−P(B)
(D) P(A∩B) = P(A)−P(B)

Qxii. $^{10}C_1 + {}^{10}C_3 + {}^{10}C_5$ = _____

(A) 380
(B) 382
(C) 384
(D) 386

Qxiii. The expansion of $(x + \sqrt{(x^2-1)})^5 + (x - \sqrt{(x^2-1)})^5$ is a polynomial of degree

(A) 3
(B) 4
(C) 5
(D) 6

Qxiv. The domain of $f(x) = \sqrt{((2-x)/(x+2))}$ is _____

(A) [−4,−2)
(B) [−4,−2]−{1}
(C) (−2,2)
(D) (−2,2]

Qxv. $(\sec\theta + \tan\theta)(1-\sin\theta)$ = _____

(A) $\sin^2\theta$
(B) $\cos^2\theta$
(C) $\mathrm{Tan}^2\theta$
(D) $\cos\theta$

Qxvi. $\sin(\sin^2\theta - \pi/2)$ = _____

(A) $\sin\theta$
(B) $-\sin\theta$
(C) $\cos\theta$
(D) $-\cos\theta$

Qxvii. $\cos \alpha =$ _____

(A) $(b^2+c^2-a^2)/2bc$ (B) $(a^2+b^2-c^2)/2bc$

(C) $(a^2+c^2-b^2)/2bc$ (D) $(b^2+c^2-a)/2bc$

Qxviii. If $\alpha = 15, b=80, y=38°$ then the area of triangle is equal to

(A) $(ab/2) \sin \alpha$ (B) $(ab/2) \sin \beta$

(C) $(ab/2) \sin \gamma$ (D) $(ab/2) \cos \alpha$

Qxix. $\text{Tan}^{-1}(x/y) - \text{Tan}^{-1}((x-y)/(x+y)) =$ _____

(A) $\pi/2$ (B) $\pi/3$

(C) $\pi/4$ (D) $3\pi/4$

Qxx. The maximum value of $y = 1 + 2 \sin \theta$ is _____

(A) 3 (B) -1

(C) 1 (D) 3

SECTION – B

Time Allowed: 2:40 Hrs | **Maximum Marks:** 50

Q2. Answer any TEN of the following parts. All questions carry equal marks:

i) If $z_1 = 1 + 2i$ and $z = 2+3i$ then evaluate $|z_1, z_2|$

ii) Prove that $|1\ a\ a^2; 1\ b\ b^2; 1\ c\ c^2| = (a - b)(b - c)(c - a)$

iii) Find the area of a parallelogram whose diagonal area $\vec{a} = 3i + 2j - 2k$ and $\vec{b} = I - 3j + 4k$.

iv) the sum of three numbers in an arithmetic sequence is 36 and the sum of their cubes is 6336. Find them.

v) The sum of three number in G.P is 38, and their product is 1728. Find them.

vi) Find the sum $1.2 + 2.3 + 3.4 + ...+ 99.100$.

vii) Find n, when ${}^{2n}C_3 : {}^{n}C_2 = 36 : 3$

viii) Find the middle term in the expansion of $(3x - x^2/2)^9$

ix) Prove that $(\sin \alpha+\sin 9\alpha)/(\cos \alpha+\cos 9\alpha) = \tan 5a$

x) the angle of depression of a boat 65.7 m from the base of a cliff is 28.9°. How high is the cliff?

xi) if $a = 92$, $b = 71$, $y = 56° 44'$. Find the area of triangle.

xii) Prove that $\text{Sin}^{-1} x + \text{Cos}^{-1} x = \pi/2$

xiii) Find the inverse of $f(x) = 2x - 3$

SECTION – C

Maximum Marks: 30

Note: Note: Attempt any Three Questions. All questions carry equal marks:

Q.3 (a) Solve the matrix method: $x + y + z = 1$; $x + y - 2z = 3$; $2x + y + z = 2$

(b) Find the volume of the parallelepiped whose edges are represented by: $a = 3i + j - k$, $b = 2i - 3j + k$, $c = I - 3j - 4k$

Q.4 (a) For what value of n, $(a^{n+1}+b^{n+1})/(a^n+b^n)$ is the arithmetic mean between a and b?

(b) Find n and r if ${}^nP_r = 840$ and ${}^nC_r = 35$

Q.5 (a) Prove by mathematical induction: $3 + 7 + 11 + ...+ (4n - 1) = n(2n + 1)/1$

(b) Find the maximum and minimum of the function $1/(19 - 10 \sin(3\theta - 45))$

Q.6 (a) Prove that $\sin 2\theta - 4 \sin^3\theta \cos\theta = \sin 2\theta \cos 2\theta$

(b) Find the area of the inscribed circle of the triangle with measure of the sides 55m, 25m and 70m